Analysis 10: A Focused Simulation Check

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1. Expected crossing delay

We check Analysis 10 without changing its theory. The model is $Y_i = \theta + \xi_i$, with independent $\xi_i \sim N(0,1)$; the sample-mean and zero predictors use squared loss and $w_i = i^\alpha$, with validation times $i = 2, \ldots, n+1$. The first two checks are exact deterministic calculations. The third uses actual Gaussian-path Monte Carlo.

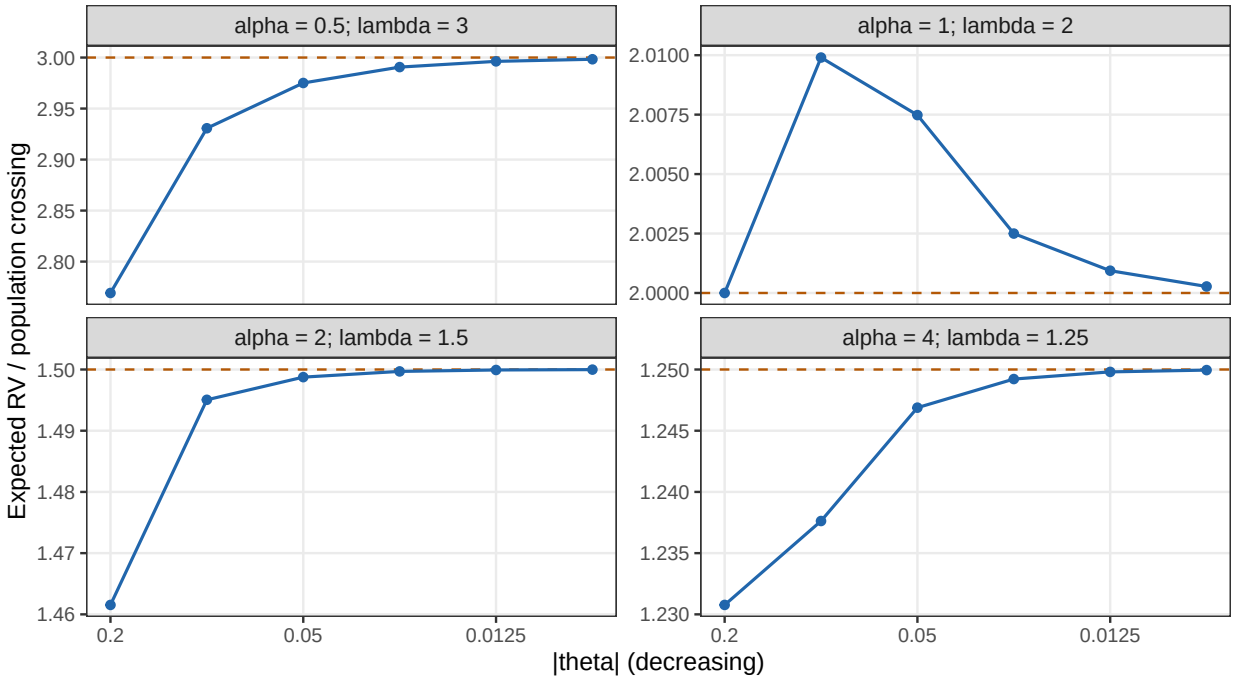


Figure 1. Exact first-strict-crossing ratios (blue) and $\lambda = 1+1/\alpha$ (dashed orange). The logarithmic $|\theta|$ axis decreases to the right. The grid is $0.2, 0.1, 0.05, 0.025, 0.0125, 0.00625$.

Interpretation. The ratios approach their fixed-delay targets. At the smallest $|\theta|$, they are 2.9983, 2.0003, 1.5000, 1.2500 for $\alpha = 0.5, 1, 2, 4$, respectively, versus $3, 2, 1.5, 1.25$. Integer rounding causes small deviations; convergence need not be monotone. These are expected-score crossings, not realized selection rates.

2. Averaging strength

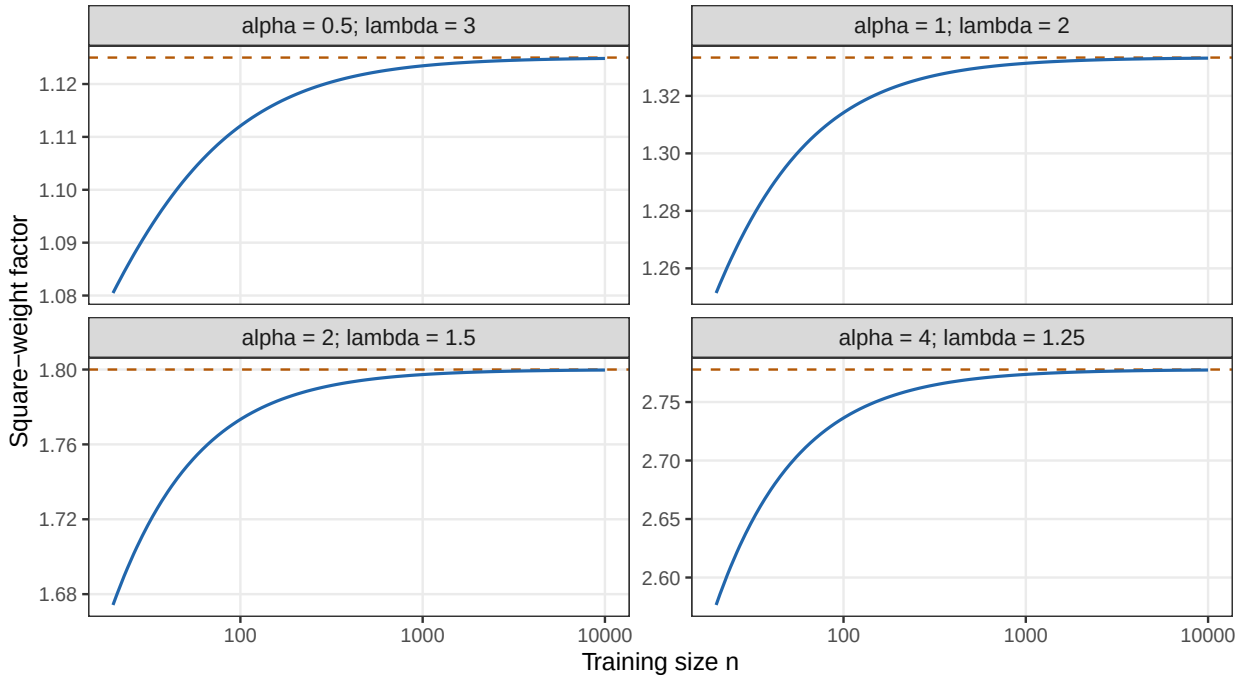


Figure 2. Exact $F_n = n\sum_{i=2}^{n+1} w_i^2/(\sum_{i=2}^{n+1} w_i)^2$ (blue) against $(\alpha+1)^2/(2\alpha+1) = \lambda^2/(\lambda^2-1)$ (dashed orange), for training sizes from 20 to 10,000.

Interpretation. At $n = 10{,}000$, the factors are 1.1248, 1.3331, 1.7997, 2.7774, respectively. Larger α gives a smaller crossing delay but worse averaging. Each fixed $\lambda > 1$ has a finite limit, unlike the forced divergence under exact alignment in Analysis 09. This factor measures weight concentration only; it is not the full Chebyshev ratio and does not by itself establish concentration.

3. Full moving-crossing Chebyshev ratio

For $\alpha = 1$ and $\lambda = 2$, use $n = 200, 500, 1000, 2000$ and $c = 0.5, 1, 1.5, 1.8, 2.2, 2.5, 3$, setting $\theta = \sqrt{c/n}$. Each configuration uses 5,000 independent Gaussian paths generated in R (seed 20261014), reused across n and c. The empirical ratio uses the sample variance divided by the squared sample mean of $\Xi_{n,w}$.

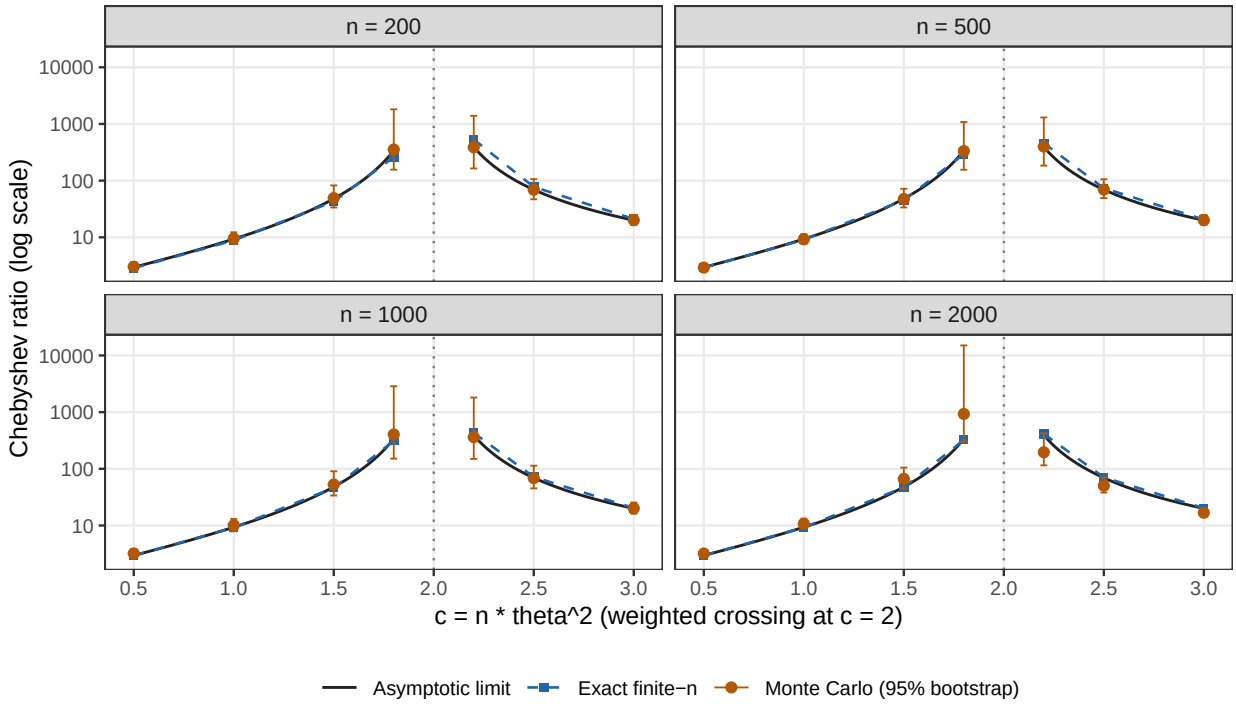


Figure 3. Empirical ratios and the Analysis 10 limit $R(2, c) = (4 + 16c/3)/(2 - c)^2$. Exact finite-$n$ ratios use the Gaussian variance identity. Bars are pointwise percentile intervals from 400 resamples of the simulated paths. Configurations are paired, not independent across panels. No values are joined across $c = 2$.

Interpretation. At $n = 2000$, exact finite-n ratios are within 5.3% of the limit on this grid. Monte Carlo ratios at $|c - 2| \geq 0.5$ differ from exact values by at most 41.5%. Near the weighted crossing, $c = 1.8$: empirical 930.0, exact 323.0, limit 340.0; $c = 2.2$: empirical 195.6, exact 414.3, limit 393.3. The sample-mean standard error is as much as 28.9% of the exact mean there, making the squared denominator noisy even with 5,000 paths.

The crossing and averaging checks closely support the predicted tradeoff. Exact finite-n full ratios approach positive limits, and the Monte Carlo check must be read with its uncertainty near $c = 2$. This does not establish a vanishing ratio or a limiting selection-error probability. Like Analysis 09, results are cached as an RDS file; re-rendering draws the figures from numerical results, not saved images. Code or configuration changes invalidate the cache automatically.